Solution Architecture

Date	27 June 2026
Team ID	SWTID-2026-5060
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out the Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components of the Credit Card Approval Prediction System, showing how applicant information flows through the web interface, is processed by the Flask application and machine learning model, and is stored for prediction history and reporting. The architecture should clearly illustrate the interaction between the user interface, application logic, machine learning components, and data storage to provide fast, accurate, and reliable credit card approval predictions.

Presentation / Client Layer

Flask Web Application
(User Interface / Dashboard)

↓↑

API Gateway / Backend Layer

Flask Web Server
(Request Handling)

↓↑

Auth Service	Core Logic Service	External API Integrations
No Authentication *(Current Version)*	*Input Validation* *Data Preprocessing* *Feature Engineering* *Random Forest Prediction* *Risk Audit Report Generation*	*Random Forest Model* *Scikit-learn* *Pandas* *NumPy*

↓↑

Data / Storage Layer

prediction_summary.json

Random Forest Model (.joblib)

Local JSON Storage for Prediction History and Trained Machine Learning Model

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides user interface for entering event details, interests, viewing conversation suggestions, fact checking, and downloading reports	Streamlit, HTML, CSS
API Gateway	Handles incoming API requests, validates data, and routes requests to appropriate services	FastAPI, Uvicorn
Microservices	Performs event analysis, topic extraction, conversation generation, history logging, feedback collection, and fact verification	Python, Transformers, Torch, Wikipedia-API
Database / Storage	Stores conversation history and user feedback for future access and analysis	JSON Files (`history.json`, `feedback.json`)
External API Integration	Retrieves verified information and NLPbased text analysis	Wikipedia API, Hugging Face Transformers
Testing Layer	Validates API endpoints and service functionality	PyTest